

Spatial-Temporal High-Frequency Learning for Video-based Visible-Infrared Person Re-Identification

Sichen Tao, Shuang Li, Jun Ye, Neng Dong, Fan Li, and Huafeng Li

Abstract—Video-based Visible-Infrared Person Re-Identification (VVI-ReID) aims to learn consistent person feature representations across video sequences in different modalities. Existing methods that use an intermediate modality to bridge the gap between visible (RGB) and infrared (IR) sequences tend to be limited by high construction costs, loss of high-frequency details, and lack of temporal cues. Moreover, they typically focus on refining global representation using high-level features, neglecting the enhancement of local details through low-level features. To address these challenges, we propose the novel Spatial-Temporal High-Frequency Learning (STHF) framework, which constructs an appropriate intermediate modality for the VVI-ReID task and alleviates the modality gap via hierarchical feature enhancement. Specifically, we introduce the Spatial-Temporal High-Pass Filter (ST-HPF), which filters out spatial-temporal Low-Frequency Components (LFC), preserving high-frequency details to construct an intermediate modality at the sequence level. We then enhance the local details with low-level features through the Shallow Detail Compensation (SDC) module, which reduces local noise interference. Finally, the Deep Semantic Refinement (DSR) module refines the global representation by modeling spatial-temporal high-frequency semantic associations using high-level features. Extensive experiments demonstrate that our method significantly outperforms state-of-the-art approaches on the publicly available HITSZ-VCV and BUPTCampus datasets. The code is available at <https://github.com/TSC95720/STHF>.

Index Terms—Video-Based Visible-Infrared Person Re-Identification, Spatial-Temporal High-Frequency Information, Sequence-Level Intermediate Modality

I. INTRODUCTION

PERSON Re-identification (ReID) involves matching the identities of pedestrians across different camera views. It has become a prominent research area due to its substantial

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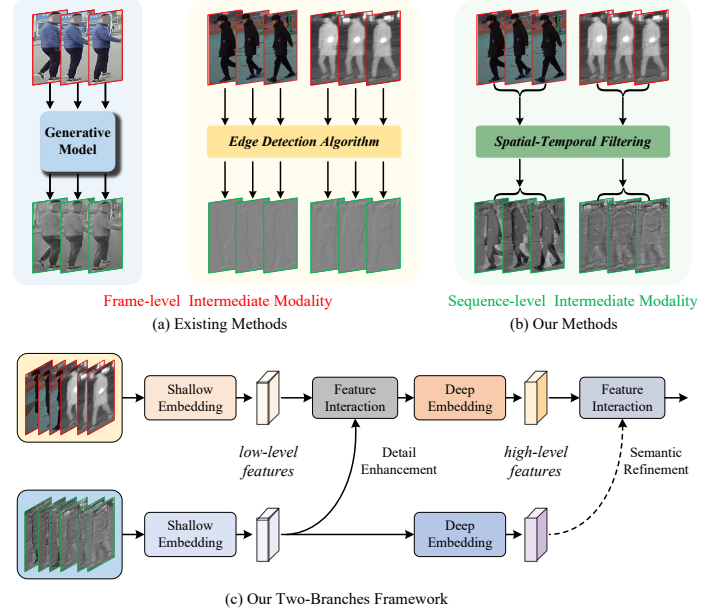


Fig. 1. Comparison with existing intermediate modality-based methods and the diagram of our method. (a) Existing VVI-ReID methods involve high costs in constructing intermediate modality (i.e., Fake IR [8]), while also suffering from the loss of high-frequency detailed information in the generated modality (i.e., Anaglyph [9]). Furthermore, these methods rely solely on single-frame images when constructing intermediate modalities, neglecting temporal modeling. (b) Our method leverages 3D FFT to extract discriminative spatial-temporal high-frequency details, constructing a sequence-level intermediate modality. (c) Our method utilizes the intermediate modality to enhance the local details of the original modality at shallow layers while guiding the extraction of semantic information at deep layers.

potential in intelligent security and video surveillance [1]–[6]. Although Person ReID has made considerable progress, it faces inherent limitations. Primarily, performance degrades significantly in challenging environments, such as low-light or night-time conditions, due to compromised image quality. Moreover, image-based ReID relying solely on single-frame images makes it unsuitable for real-world applications. Consequently, Video-based Visible-Infrared Person Re-Identification (VVI-ReID) [7], [8] has emerged as a promising solution, aiming to match pedestrian sequences captured by night-time infrared (IR) cameras with those captured by day-time visible (RGB) cameras.

The core task of VVI-ReID is to bridge the modality gap between IR and RGB pedestrian sequences. Most existing methods [7], [10], [11] attempt to mitigate the modality

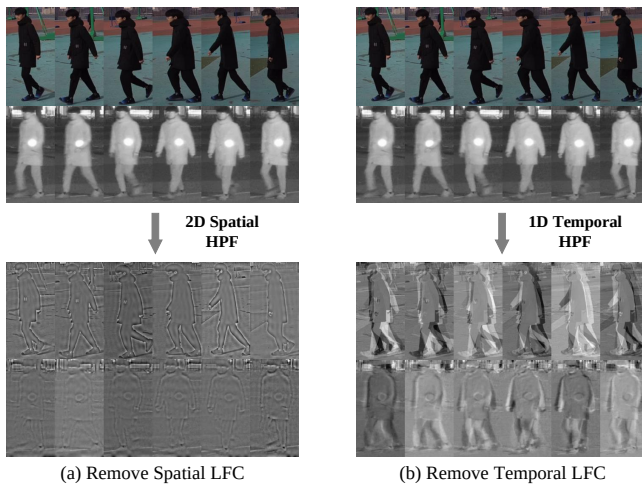


Fig. 2. The influence of removing the Low-Frequency Components (LFC) in video pedestrian sequences. Where the HPF denotes the ideal high-pass filter. (a) Filtering spatial LFC can reduce spatial redundant information such as background, while retaining fine-grained structural details of pedestrians. (b) Removing temporal LFC can filter out static temporal information while capturing the motion patterns of pedestrians.

discrepancy by learning shared feature embeddings. However, due to fundamental differences in the imaging mechanisms of the RGB and IR sequences, substantial variations in color, texture, and contrast persist, making direct feature alignment ineffective and susceptible to identity information loss. To address this issue, intermediate modality-based methods [8], [9] generate an intermediary modality to bridge the gap between RGB and IR sequences, enhancing feature alignment while preserving identity information. Despite their promising performance, these methods still face notable limitations. **Specifically, as shown in Fig. 1(a), the construction of intermediate modalities is inappropriate due to the following reasons:** 1) **High construction cost:** generation-based method [8] demand substantial computational resources and are inherently prone to mode collapse [12], [13]; 2) **Loss of high-frequency detail:** edge detection-based method [9] typically focus on capturing abrupt intensity changes and may smooth out subtle variations, leading to the loss of high-frequency detail; 3) **Lack of temporal cues:** these methods relying on single-frame construct intermediate modalities, neglecting that the temporal domain also contains important discriminative information, such as temporal high-frequency information. **In addition, the utilization of intermediate modalities is insufficient:** Deep feature learning follows a low-level to high-level paradigm, as shown in Fig. 1(c), yet previous methods [8], [9] typically use the high-level semantic features of the intermediate modality to refine the global representation while neglecting the low-level features that are crucial for capturing fine-grained local details. These factors limit intermediate modality-based methods from effectively alleviating the modality discrepancy and learning discriminative feature representations.

Recently, transforming signals into the frequency domain using the Fast Fourier Transform (FFT) has been widely adopted in deep learning [14]–[17]. Compared with the spatial domain, the discriminability of features is physically defined

in the frequency domain. As illustrated in Fig. 2, by removing the Low-Frequency Components (LFC) from video pedestrian sequences, the spatial High-Frequency Components (HFC) preserve fine-grained structural details, while temporal HFC capture motion variations, both of which provide distinct modality-invariant discriminative cues. Furthermore, FFT has a low computational complexity of $O(N \log N)$, making it significantly more efficient in processing large-scale video data compared to traditional spatial domain methods. Thus, as shown in Fig. 1(b), learning spatial-temporal discriminative information from a frequency-domain perspective is crucial for the VVI-ReID task.

Based on the above analysis, in this paper, we propose a Spatial-Temporal High-Frequency Learning (STHF) framework for the VVI-ReID task. The proposed STHF aims to mitigate the modality discrepancy while learning spatial-temporal high-frequency information from low-level to high-level features. To achieve this, we propose three key modules: Spatial-Temporal High-Pass Filter (ST-HPF), Shallow Detail Compensation (SDC) module, and Deep Semantic Refinement (DSR) module. The proposed ST-HPF constructs a novel sequence-level intermediate modality based on a 3D Fast Fourier Transform (FFT) by explicitly leveraging spatial-temporal cues to enhance identity discriminability. Based on this, the proposed SDC module enhances local structural details with the low-level features by accurately modeling spatial correspondence between the intermediate modality and the original modality, while the proposed DSR module refines global representations with the high-level features by establishing spatial-temporal semantic associations. To ensure effective information interaction, we first eliminate the style information of the original modality before applying the SDC and DSR modules. By learning spatial-temporal high-frequency information, STHF effectively mitigates the modality discrepancy and extracts discriminative spatial-temporal representations.

Our main contributions can be summarized as follows:

- We propose a novel Spatial-Temporal High-Frequency Learning (STHF) framework to exploit the potential spatial-temporal high-frequency information for the VVI-ReID task.
- We propose a Spatial-Temporal High-Pass Filter (ST-HPF) based on 3D FFT to construct a novel sequence-level intermediate modality, which jointly addresses modality alignment and spatial-temporal feature learning.
- We propose a Shallow Detail Compensation (SDC) module to enhance local details with low-level features, and a Deep Semantic Refinement (DSR) module to refine global representations with high-level features.
- Experiments on HITSZ-VCM and BUPTCampus demonstrate that our method significantly outperforms state-of-the-art approaches, validating the effectiveness of the proposed ST-HPF, SDC, and DSR.

The remainder of this paper is organized as follows. Section II reviews related work; Section III presents the proposed method; Section IV analyzes the experimental results; Finally, section V concludes the paper.

II. RELATED WORK

A. Visible-Infrared Person Re-Identification

Visible-Infrared Person Re-Identification (VI-ReID) is a cross-modality person retrieval task that aims to match individuals across cameras with different modalities. To alleviate the discrepancy between RGB and IR images, numerous works [18]–[23] have been proposed and have achieved remarkable performance. The current mainstream methods can be categorized into three types: representation learning-based methods [24]–[28], metric learning-based methods [29]–[32], and generation-based methods [33]–[37].

Representation learning-based methods primarily focus on extracting shared and discriminative features across the two modalities by designing and modeling appropriate network architectures. IDKL [24] leverages the discriminative knowledge embedded in modality-specific features to enhance the discriminability of modality-shared features. MSCMNet [25] proposes to fuse features at different scales and explores the semantic correlation of fusion. DMANet [26] develops an effective multi-granularity features mutual learning module to eliminate the modality discrepancy. DEEN [27] learns informative feature representations by generating diverse embeddings while reducing the modality discrepancy between RGB and IR images. DMA [28] proposes to perform compensation for the information asymmetry in the HSV color space.

Metric learning-based methods aim to reduce the distance between samples of the same identity across different modalities by designing appropriate feature metrics or loss functions. HHGF [29] proposes the CMCC loss function to mine the invariance of global features by measuring mutual information in images from two modalities. SDCL [30] effectively mitigates the cross-modality discrepancy through the collaboration of shallow and deep features. HOS-Net [31] proposes a modality-range identity-center contrastive loss to reduce the distances between the RGB, IR, and intermediate features. CPM [32] introduces the closest permutation distance that is invariant to changes in the order of the group members to measure the similarity between two sets of features.

Generation-based methods generally transform cross-modality tasks into single-modality tasks through modality conversion or seek an intermediate modality between the two modalities to alleviate the modality discrepancy. GC-IFS [33] generates high-quality cross-modality pairs of images and fuses the information of the two modalities. PCMC [34] splits the two modalities of the same person into patches and concatenates them into a new modality image, effectively alleviating the problem of modality imbalance. HAT [35] proposes an auxiliary grayscale modality generated from homogeneous RGB, which preserves the structural information of visible images while approximating the image style of the infrared modality. XIV [36] employs a lightweight network trained in a self-supervised manner to generate an X modality.

Although the above methods have achieved notable success, the image-based VI-ReID is suboptimal for video sequence retrieval due to its limited ability to capture temporal features, resulting in inherent information loss.

B. Video-based Visible-Infrared Person Re-Identification

Compared to single-frame images, video sequences provide richer spatial content and implicit temporal cues, which have attracted increasing attention to VVI-ReID in recent years. Inspired by previous works in video action recognition [38]–[40] and video-based person re-identification [41]–[49], existing VVI-ReID methods often introduce additional networks such as Recurrent Neural Networks (RNNs) [7], [9], Graph Neural Networks (GNNs) [10], and Transformers [50], [51] to learn discriminative spatial-temporal representations and thus alleviate the modality discrepancy between RGB and IR video sequences.

RNNs-based methods leverage recurrent structures to model sequential dependencies. For instance, MITML [7] first introduces the VVI-ReID task, contributes the HITZS-VCM dataset, and employs an LSTM-based [52] temporal memory module to aggregate frame-level features. IBAN [9] proposes a bidirectional LSTM to integrate temporal features across frames and leverages the Anaglyph intermediate modality to alleviate the modality discrepancy. In contrast, **GNNs-based methods**, such as SAADG [10] formulates modality discrepancies as style attacks and applied a Graph Neural Network [53] to extract robust cross-modality representations by modeling intra- and inter-modality relations. Recently, **Transformers-based methods** have demonstrated great potential in VVI-ReID. CST [50] proposes a cross-modality spatial-temporal Transformer [54] that encapsulates local pedestrian information into 3D tubes and facilitates inter-frame interactions via message tokens. STIMM [51] employs a temporal Transformer to capture comprehensive motion patterns, thereby enhancing feature discriminability across frames. Other methods, such as AuxNet [8], proposes a temporal k-reciprocal re-ranking strategy to enhance feature matching over time.

These methods have achieved significant success, fully demonstrating the feasibility and effectiveness of the VVI-ReID task. However, they primarily focus on mitigating the modality discrepancy and extracting spatial-temporal information in the spatial domain, while overlooking potential solutions in the frequency domain.

C. Fourier Transform

The Fourier Transform has been increasingly applied in deep learning for its ability to model global dependencies in the frequency domain. Recent studies have further demonstrated its effectiveness across a range of tasks [14]–[16], [55].

For example, in medical image segmentation, FreMIM [14] leverages Fast Fourier Transform to replace self-attention, enabling efficient global information modeling. In human pose estimation, FTCLM [15] models the frequency and temporal interactions between poses through separate feature mixing operations. In the area of low-light image enhancement, FourLIE [56] explores the positive correlation between amplitude magnitude and brightness magnitude. For high-quality image deblurring, FFTformer [57] develops an efficient frequency-based self-attention solver, which reduces spatial and temporal complexity while improving efficiency and effectiveness.

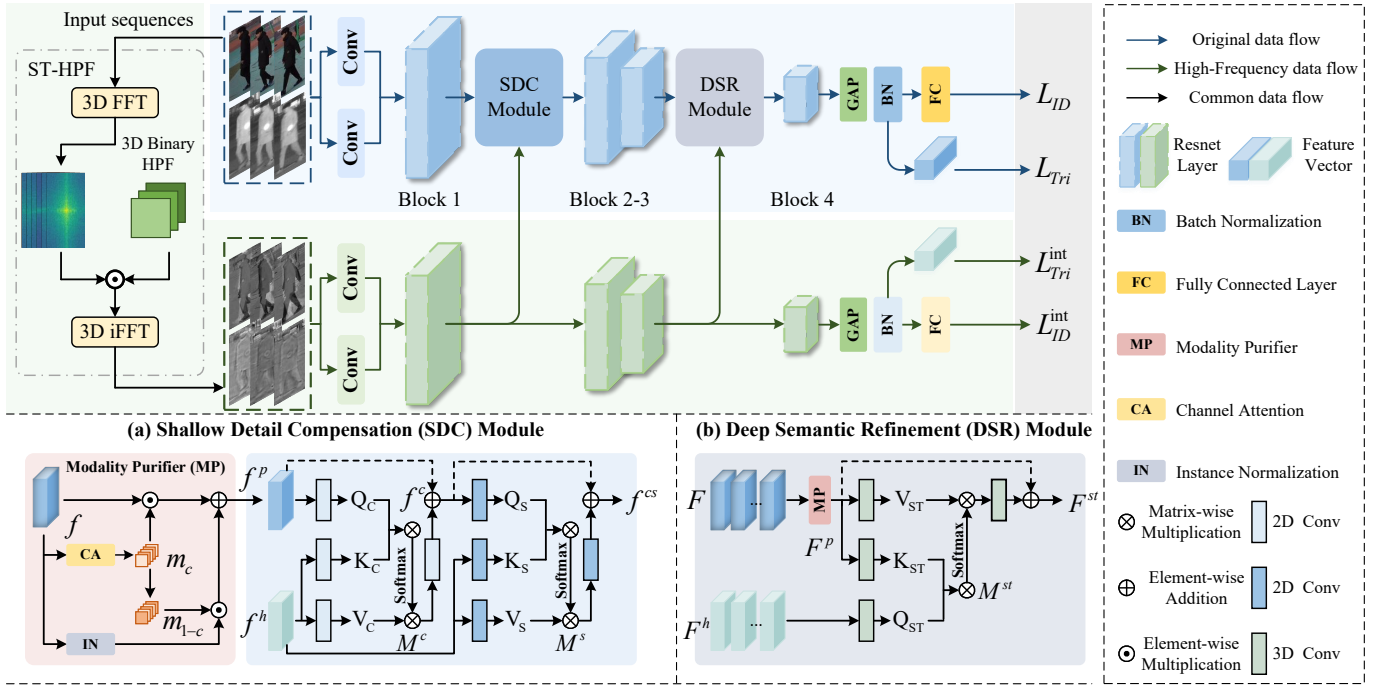


Fig. 3. The framework of STHF, consists of a Spatial-Temporal High-Pass Filter (ST-HPF), a Shallow Detail Compensation (SDC) module, and a Deep Semantic Refinement (DSR) module. The ST-HPF first constructs a sequence-level intermediate modality through spatial-temporal high-pass filtering. Then SDC compensates the original modality with high-frequency spatial information in the shallow layers of ResNet, while DSR guides the original modality to extract spatial-temporal high-frequency information in the deep layers. The two information interaction modules operate under modality-irrelevant conditions by the modality purifier. We perform SDC operations in blocks 1 – 2 of ResNet and DSR operations in blocks 3 – 4. For simplicity, only block 1 and block 3 are depicted in the figure to exemplify the integration of SDC and DSR modules.

In face super-resolution, SFMNet [58] designs a spatial-frequency mutual learning network, where the frequency domain branch captures global dependency and the spatial domain branch extracts local dependency. The Fourier Transform has also been widely adopted in person ReID. For instance, FDMNet [59] learns modality-invariant features by enhancing modality-invariant amplitude components and suppressing modality-specific components. FDNM [60] improves the discriminability of phase components by leveraging amplitude guidance in the frequency domain, effectively learning robust feature representations. FIDN [61] disentangles and utilizes low-frequency and high-frequency spatial-temporal to capture comprehensive and discriminative video representations.

Inspired by the success of these works, we propose a spatial-temporal high-pass filtering method for VVI-ReID to exploit the potential spatial-temporal high-frequency information in pedestrian sequences.

III. PROPOSED METHOD

A. Basics of Fast Fourier Transform

The core task of VVI-ReID is to bridge the modality gap between RGB and IR pedestrian sequences. Previous intermediate modality-based methods [8], [9] focused mainly on constructing intermediate modalities in the spatial domain to mitigate the modality discrepancy, while neglecting the discriminative spatial-temporal high-frequency information, which is crucial for the VVI-ReID task. Therefore, before introducing STHF, we first review the basics of the Fast

Fourier Transform (FFT). Specifically, Given a 1D signal $x[n]$ of length N , then the 1D FFT is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(\frac{2\pi}{N})kn}, \quad (1)$$

where j is the imaginary unit. $X[k]$ denotes the k -th frequency component. Since the FFT is a linear transformation, the original signal can be reconstructed using the inverse Fast Fourier Transform (iFFT):

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(\frac{2\pi}{N})kn}. \quad (2)$$

The 1D-FFT can be naturally extended to its multidimensional FFT by sequentially applying 1D FFT along each dimension, which can be defined as:

$$X_{\mathbf{k}} = \sum_{\mathbf{n}=0}^{\mathbf{N}-1} x_{\mathbf{n}} e^{-j2\pi\mathbf{k}(\frac{\mathbf{n}}{\mathbf{N}})}, \quad (3)$$

where $\mathbf{k} = (k_1, k_2, \dots, k_d)$ and $\mathbf{n} = (n_1, n_2, \dots, n_d)$ are the frequency and spatial indices in d dimensions, respectively, and $\mathbf{N} = (N_1, N_2, \dots, N_d)$ denotes the size of each dimension. For conciseness, the expression of the d -dimensional iFFT, which can be directly derived from Eq. (2), is not explicitly presented.

In the frequency domain, both spatial and temporal HFC contain unique modality-invariant discriminative information (e.g., edges, motion artifacts). Thus, by filtering out the spatial-temporal LFC from the RGB and IR pedestrian sequences,

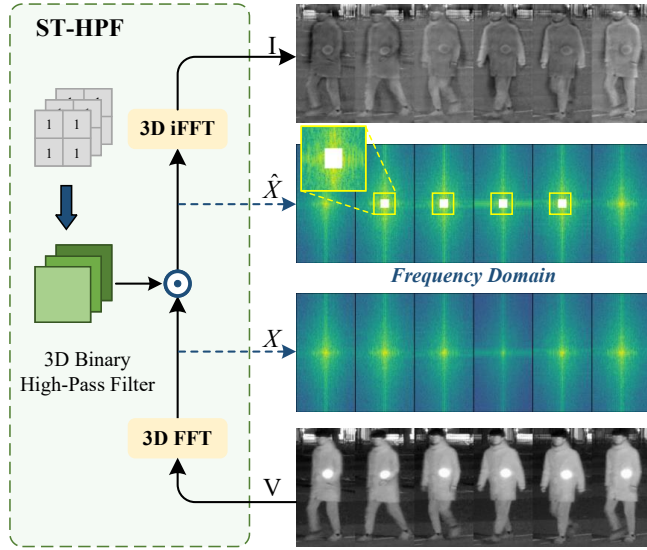


Fig. 4. Spatial-Temporal High-Pass Filter (ST-HPF). The ST-HPF first maps the original pedestrian sequence from the spatial domain to the frequency domain using 3D FFT. Then, a 3D ideal high-pass filter is applied to remove the LFC from the sequence. Finally, the sequence spectrum is mapped back to the spatial domain by 3D iFFT, obtaining a spatial-temporal high-frequency pedestrian sequence. Figures on the right are an example of applying ST-HPF. In the frequency domain, the spatial-temporal LFC (small white regions inside yellow boxes) are removed.

identity-related features that are robust to modality variations can be effectively extracted, enabling the construction of a suitable intermediate modality for the VVI-ReID task. We will elaborate on how to filter out low-frequency information while preserving useful spatial-temporal high-frequency information in ST-HPF.

B. Overview

As shown in Fig. 3, the proposed STHF mainly includes a Spatial-Temporal High-Pass Filter (ST-HPF), a Shallow Detail Compensation (SDC) module, and a Deep Semantic Refinement (DSR) module. The STHF first utilizes the ST-HPF to generate spatial-temporal high-frequency pedestrian sequences as an intermediate modality. Based on this, the SDC module enhances local details of low-level features, and the DSR module refines the global representation with high-level features.

C. Spatial-Temporal High-Pass Filter

To remove the redundant LFC across both spatial and temporal dimensions, thereby constructing a suitable intermediate modality for the VVI-ReID task, inspired by [62], we introduce a Spatial-Temporal High-Pass Filter (ST-HPF) based on 3D-FFT with the binary filtering strategy. Since the video pedestrian sequence V has three dimensions T (temporal), H (height), and W (width), the filter is also 3D with three cutoff frequencies. In the centered 3D spectrum, low spatial frequencies lie near $(k_h, k_w) = (0, 0)$, and low temporal frequencies are near $k_t = 0$, as demonstrated in [63], [64]. Following this prior, we preset the cutoff frequencies according to the sequence size. Specifically, a spatial cutoff frequency f_{co}^s handles both H and

Algorithm 1: Process of proposed ST-HPF

Input: Pedestrian video sequence V of the original modality; spatial cutoff frequency f_{co}^s ; temporal cutoff frequency f_{co}^t .

Output: Filtered sequence I with enhanced spatial-temporal high-frequency components.

Step.1: # 3D Fast Fourier Transform;

$X = FFT(V)$, $X = FFTShift(X)$;

Step.2: # Generate 3D Binary High-Pass Filter;

$F = \mathbf{1}_{T \times H \times W}$;

$t_{crop} = \frac{T}{2}$, $h_{crop} = \frac{H}{2}$, $w_{crop} = \frac{W}{2}$;

$t_{range} = (t_{crop} - f_{co}^t) : (t_{crop} + f_{co}^t)$;

$h_{range} = (h_{crop} - f_{co}^s) : (h_{crop} + f_{co}^s)$;

$w_{range} = (w_{crop} - f_{co}^s) : (w_{crop} + f_{co}^s)$;

$F[t_{range}, h_{range}, w_{range}] = 0$;

Step.3: # Spatial-Temporal High-Pass Filtering;

$\hat{X} = F \odot X$;

Step.4: # 3D inverse Fast Fourier Transform;

$\hat{X} = iFFTShift(\hat{X})$, $I = iFFT(\hat{X})$;

W dimension, while a temporal cutoff frequency f_{co}^t handles T dimension.

To achieve this, as shown in Fig. 4, we first perform the frequency domain projection to map the pedestrian sequence to the frequency domain. Then, we perform the high-pass filtering in the frequency domain to remove redundant LFC of the pedestrian sequence. Finally, we conduct the spatial domain reconstruction to convert the spectrum back to the spatial domain, obtaining the high-frequency pedestrian sequence.

Frequency Domain Projection. We first apply 3D FFT to the pedestrian sequence V , transforming it into a spectrum. Afterward, To facilitate spectrum analysis and filtering operations, we center the spectrum to ensure that the LFC is at the center. The spectrum X is obtained as follows:

$$X = FFTShift(F(V)), \quad (4)$$

where $F(\cdot)$ represents the 3D FFT, whose mathematical definition can be directly derived from Eq. (3), and $FFTShift(\cdot)$ denotes the operation that centers the frequency spectrum by shifting the LFC to the center.

High-Pass Filtering. In the frequency domain, we introduce a 3D high-pass filter F , which removes redundant LFC across both spatial and temporal dimensions through element-wise multiplication with the input sequence X . Therefore, the resulting filtered spectrum $\hat{X}$ can be obtained as follows:

$$\hat{X} = F \odot X, \quad (5)$$

where F is a binary matrix where 1 represents HFC and 0 represents LFC, constructed by retaining frequency components whose distance from the spectrum center exceeds the predefined cutoff values while suppressing those below cutoff

values. The detailed construction is as follows:

$$\mathbf{F}(t, h, w) = \begin{cases} 1, & \text{if } |t - t_{\text{crop}}| > f_{\text{co}}^t, \\ & |h - h_{\text{crop}}| > f_{\text{co}}^s, \\ & |w - w_{\text{crop}}| > f_{\text{co}}^s, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

where f_{co}^t and f_{co}^s denote the cutoff frequencies in temporal and spatial dimensions, respectively. Meanwhile, $h_{\text{crop}} = \frac{H}{2}$, $w_{\text{crop}} = \frac{W}{2}$, and $t_{\text{crop}} = \frac{T}{2}$ are the center coordinates used to determine the center position of the frequency spectrum region.

Spatial Domain Reconstruction. After the spatial-temporal high-pass filtering in the frequency domain, we restore the center position of the filtered spectrum and then map it back to the spatial domain to obtain the high-frequency pedestrian sequence I by iFFT. Which can be described as:

$$I = F^{-1}(\text{iFFTShift}(\hat{X})), \quad (7)$$

where $F^{-1}(\cdot)$ denotes the 3D iFFT, whose mathematical definition can be derived from Eq. (2) and Eq. (3). $\text{iFFTShift}(\cdot)$ denotes the operation that reverses the frequency spectrum shift. Algorithm 1 describes the procedure of ST-HPF in detail. As shown in Fig. 4, by performing the above operations, spatial-temporal low-frequency information in the pedestrian sequence is suppressed, thereby enhancing both structural and motion-based identity cues.

D. Shallow Detail Compensation (SDC) Module

Low-level features typically contain rich details by preserving textures, edges, and local patterns, but are susceptible to background interference and modality-specific noise. Therefore, we propose a Shallow Detail Compensation (SDC) module that aims to enhance the high-frequency detail of original modality with low-level features. The structure of the proposed SDC is shown in Fig. 3(a), considering the interference of style information in the original modality, we first perform modality purify on the RGB and IR images. Afterward, we sequentially perform fine-grained high-frequency channel and spatial information compensation to enhance the local details of original modality.

Modality Purify. To prevent style information from interfering with interaction, inspired by [23], we introduce a Modality Purifier (MP) before performing SCD. It eliminates style information through an Instance Normalization (IN) layer while preserving identity-related information using channel attention. Specifically, given a low-level feature maps $f \in \mathbb{R}^{C \times H \times W}$, through MP, we obtain the purified feature maps f^p :

$$f^p = m_c \odot f + (1 - m_c) \odot \text{IN}(f), \quad (8)$$

where m_c represents the channel attention mask and $\text{IN}(\cdot)$ represents the IN layer. we use SE-Net [65] to generate the channel mask m_c :

$$m_c = \sigma(W_2 \delta(W_1 g(f))), \quad (9)$$

where the $g(\cdot)$ represents global average pooling, W_1 and W_2 denote the learnable parameters of two fully connected

(FC) layers, $\delta(\cdot)$ and $\sigma(\cdot)$ represent the ReLU and sigmoid activation functions, respectively.

Channel Information Compensation. To enhance the channel information, we extract the local detail affinity between the high-frequency feature maps $f^h \in \mathbb{R}^{C \times H \times W}$ and the purified feature maps f^p along the channel dimensions. Specifically, we first calculate the channel-wise similarity matrix $M^c \in \mathbb{R}^{C' \times C'}$ by matrix multiplication followed by softmax:

$$M^c = F_{\text{softmax}}(\psi_q^1(f^p) \times \psi_k^1(f^h)), \quad (10)$$

where $\psi_k^1(f^h)$ and $\psi_q^1(f^p)$ represent the key and query embeddings to capture channel relationships, which are generated through 2D 1×1 convolutional layers. Then, we combine the value embedding $\psi_v^1(f^h)$ with the M^c through a weighted sum and add it to the f^p , completing the information compensation to obtain the channel-compensated feature maps f^c :

$$f^c = f^p + \omega^c(\psi_v^1(f^h) \times M^c), \quad (11)$$

where ω^c is the 2D 1×1 convolutional layer used to restore the feature dimensions.

Spatial Information Compensation. Furthermore, to enhance the spatial information, we compute the spatial-wise similarity matrix $M^s \in \mathbb{R}^{S \times S}$ between f^c and f^h using a similar approach:

$$M^s = F_{\text{softmax}}(\psi_q^2(f^c) \times \psi_k^2(f^h)), \quad (12)$$

where $\psi_q^2(f^c)$, $\psi_k^2(f^h)$ represent the query and key embeddings to capture spatial relationships. Then, we perform spatial information compensation, which is similar to channel information compensation, to obtain the final compensated feature maps f^{cs} :

$$f^{cs} = f^c + \omega^s(\psi_v^2(f^h) \times M^s), \quad (13)$$

where ω^s is used to restore the feature dimensions and $\psi_v^2(f^h)$ represents the value embedding of the f^h . Through channel and spatial high-frequency information compensation, SDC effectively integrates the rich high-frequency local details into the original modality.

E. Deep Semantic Refinement (DSR) Module

High-level features are essential for abstract semantic understanding and global pattern recognition. To learn discriminative spatial-temporal high-frequency representations, we propose a Deep Semantic Refinement (DSR) module that aims to refine the global representation by modeling spatial-temporal high-frequency semantic associations using high-level features. The structure of the proposed DSR is shown in Fig. 3(b). Based on the previous analysis, we first perform modality purify as well. Then, we perform spatial-temporal refinement to refine the global representation.

Modality Purify. Similar to SDC, we first utilize a Modality Purifier to eliminate identity-unrelated style information from the original modality sequence feature maps $F \in \mathbb{R}^{T \times C \times H \times W}$, and the sequence feature maps after purification are denoted as F^p :

$$F^p = \text{MP}(F), \quad (14)$$

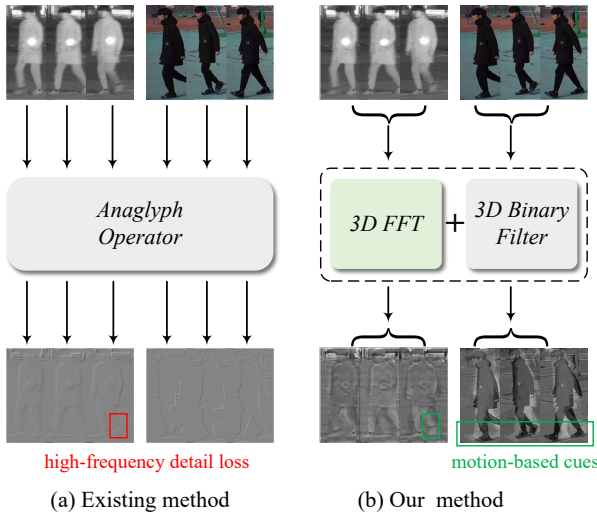


Fig. 5. Comparison of intermediate modality construction methods. (a) Existing VVI-ReID method [9] constructs the intermediate modality based on single-frame images and suffers from the loss of high-frequency detail information. (b) Our method constructs the intermediate modality at the sequence level, enhancing spatial-temporal high-frequency discriminative information.

where the operations of $MP(\cdot)$ follow the same principles as described in Eq. (8).

Spatial-Temporal Refinement. To establish global semantic associations, we extract the spatial-temporal affinity between the high-frequency sequence feature maps $F^h \in \mathbb{R}^{T \times C \times H \times W}$ and F^p . We first calculate the spatial-temporal similarity matrix $M^{st} \in \mathbb{R}^{ST \times ST}$:

$$M^{st} = F_{softmax} \left(\varphi_q(F^h) \times \varphi_k(F^p) \right), \quad (15)$$

where $\varphi_q(F^h)$ and $\varphi_k(F^p)$ represent the key and query embeddings to mine spatial-temporal associations, which are generated through 3D 1×1 convolutional layers. Next, we combine the value embedding $\varphi_v(F^p)$ with the M^{st} through a weighted sum and add it to the F^p , completing the information guidance to obtain the spatial-temporal-guided sequence feature maps F^{st} :

$$F^{st} = F^p + \omega^{st} \left(\varphi_v(F^p) \times M^{st} \right), \quad (16)$$

where ω^{st} is the 3D 1×1 convolutional layer use to restore the feature dimensions. Through spatial-temporal high-frequency information refinement, DSR guides the original modality in effectively extracting spatial-temporal information.

F. Discussion

To further highlight the uniqueness and novelty of our approach, we provide the following discussions focusing on three key aspects: intermediate modality design, hierarchical feature learning, and frequency-domain learning strategies.

Intermediate Modality Construction. Rather than constructing image-level intermediate modalities for the VVI-ReID task, STHF incorporates temporal modeling into the construction of the intermediate modality, further enhancing identity discriminability. As shown in Fig. 5(a), although the Anaglyph data [9] alleviate modality discrepancy to a large extent, they lose critical discriminative information due to gener-

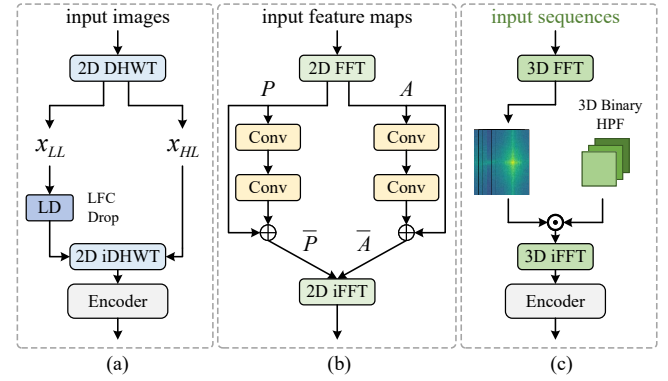


Fig. 6. Different strategies in frequency-domain learning. (a) PHA [66] first decomposes images into LFC and HFC using 2D Discrete Haar Wavelet Transform (DHWT), then drops the LFC via a Low-Frequency Drop (LD) module, and finally enhances spatial high-frequency information through representation learning. (b) FRB [58] applies 2D FFT to decompose features into amplitude and phase components, and captures image-level global dependencies via convolutional operations. (c) Our method applies 3D FFT to the input sequences and removes spatial-temporal LFC using a 3D binary high-pass filter, followed by representation learning to extract spatial-temporal high-frequency information.

ating the intermediate modality from a single-frame image, ignoring the inherent spatial-temporal correlations within video sequences. In contrast, the proposed ST-HPF constructs a sequence-level intermediate modality by removing the spatial-temporal LFC from IR and RGB pedestrian sequences, which not only reduces modality discrepancy but also enhances high-frequency details in both temporal and spatial dimensions, as illustrated in Fig. 5(b). Therefore, constructing a highly discriminative sequence-level intermediate modality is crucial for the VVI-ReID task.

Hierarchical Representation Enhancement. STHF introduces two distinct feature interaction modules to enhance representations from both shallow and deep embedding space. First, the Shallow Detail Compensation (SDC) module leverages low-level features through 2D CNNs at the frame level to compensate for local detail loss. Then, the Deep Semantic Refinement (DSR) module employs high-level features via 3D CNNs at the sequence level to refine global semantic representations. Through the combined effect of SDC and DSR, STHF learns more comprehensive and discriminative representations.

Spatial-Temporal Frequency Modeling. STHF introduces a frequency-domain strategy well-suited to the VVI-ReID task. As shown in Fig. 6, compared with capturing spatial high-frequency cues via coarse frequency separation through 2D Discrete Haar Wavelet Transform (DHWT) [66] and modeling global frequency information in the spatial domain using 2D FFT [58], STHF employs 3D FFT combined with a 3D binary filter to extract spatial-temporal high-frequency information, thus preserving the complete frequency spectrum and further enhancing identity discriminability.

G. Optimization

We combine cross-entropy loss and triplet loss to jointly optimize our network for learning discriminative and modality-

TABLE I

COMPARE WITH THE STATE-OF-THE-ART PERSON RE-ID METHODS ON HITSZ-VCM. “R@1”, “R@5” AND “R@10” DENOTE RANK-1, RANK-5, AND RANK-10, RESPECTIVELY. THESE RESULTS ARE COPIED FROM THEIR PAPERS.

Methods	Reference	Type	Seq_Len	Infrared to Visible				Visible to Infrared			
				R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP
DDAG [67]	ECCV’20	Image	6	54.6	69.8	76.0	39.3	59.0	74.6	79.5	41.5
Lba [68]	ICCV’21	Image	6	46.4	65.3	72.2	30.7	49.3	69.3	75.9	32.4
MPANet [23]	CVPR’21	Image	6	46.5	63.1	70.5	35.3	50.3	67.3	73.6	37.8
VSD [69]	CVPR’21	Image	6	54.5	70.0	76.3	41.2	57.5	73.7	79.4	43.5
CAJ [70]	ICCV’21	Image	6	56.6	73.5	79.5	41.5	60.1	74.6	79.9	42.8
SEFL [71]	CVPR’23	Image	6	67.7	80.3	84.7	52.3	70.2	82.2	86.1	52.5
MITML [7]	CVPR’22	Video	6	63.7	76.9	81.7	45.3	64.5	79.0	83.0	47.7
IBAN [9]	TCSVT’23	Video	6	65.0	78.3	83.0	48.8	69.6	81.5	85.4	51.0
SADSTRM [72]	Arxiv’23	Video	6	65.3	77.9	82.7	49.5	67.7	80.7	85.1	51.8
SAADG [10]	ACM MM’23	Video	6	69.2	80.6	85.0	53.8	73.1	83.5	86.9	56.1
CST [50]	TMM’24	Video	6	69.4	81.1	85.8	51.2	72.6	83.4	86.7	53.0
AuxNet [8]	TIFS’24	Video	6	51.1	-	-	46.0	54.6	-	-	48.7
our	-	Video	6	70.4	81.6	86.2	56.2	73.5	83.7	87.0	58.6

TABLE II

COMPARE WITH THE STATE-OF-THE-ART PERSON RE-ID METHODS ON BUPTCAMPUS. “R@1”, “R@5” AND “R@10” DENOTE RANK-1, RANK-5 AND RANK-10, RESPECTIVELY. THESE RESULTS ARE COPIED FROM THEIR PAPERS. AUXNET* INDICATES THAT WE DID NOT USE THE RE-RANKING ALGORITHM IN THIS METHOD.

Methods	Reference	Type	Seq_Len	Infrared to Visible				Visible to Infrared			
				R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP
AlignGAN [73]	ICCV’19	Image	10	28.0	49.1	57.7	30.3	35.4	53.9	61.3	35.1
DDAG [67]	ECCV’20	Image	10	40.4	61.4	69.8	40.4	46.3	68.2	74.4	43.1
LbA [68]	ICCV’21	Image	10	32.1	54.9	65.1	32.9	39.1	58.7	66.5	37.1
CAJ [70]	ICCV’21	Image	10	40.5	66.8	73.3	41.5	45.0	70.0	77.0	43.6
AGW [1]	TPAMI’21	Image	10	36.4	60.1	67.2	37.4	43.7	64.4	73.2	41.1
MMN [74]	CVPR’21	Image	10	40.9	67.2	74.4	41.7	43.7	65.2	73.5	42.8
DART [75]	CVPR’22	Image	10	52.4	70.5	77.8	49.1	53.3	75.2	81.7	50.5
DEEN [27]	CVPR’23	Image	10	53.7	74.8	80.7	50.4	49.8	71.6	81.0	48.6
MITML [7]	CVPR’22	Video	6	49.1	67.9	75.4	47.5	50.2	68.3	75.7	46.3
AuxNet* [8]	TIFS’24	Video	6	58.4	76.1	84.1	55.5	58.4	76.4	81.1	53.5
our	-	Video	6	66.7	81.8	87.0	62.1	59.8	76.8	82.8	54.6

invariant features. Therefore, the total loss of the proposed STHF is formulated as:

$$L_{total} = L_{ID} + L_{Tri} + \lambda_1 L_{ID}^{int} + \lambda_2 L_{Tri}^{int}, \quad (17)$$

where λ_1 and λ_2 are the hyper-parameters, L_{ID} is the cross-entropy loss and L_{Tri} is the triplet loss. During training, L_{ID} and L_{Tri} work on the RGB and IR data, while L_{ID}^{int} and L_{Tri}^{int} work on high-frequency data.

IV. EXPERIMENTS

A. Datasets and Evaluation Metrics

HITSZ-VCM [7] comprises 251452 RGB images and 211807 IR images from 927 identities captured across 12 cameras. Each pedestrian track is composed of 24 sequential frames, and all images are categorized into 11,785 RGB tracks and 10,078 IR tracks.

BUPTCampus [8] comprises 1869066 images of 3080 different identities, all captured from 6 camera perspectives. Each track of pedestrians consists of 111 sequential frames, and all images are categorized into 16826 tracks.

Evaluation Metrics. We use Cumulative Matching Characteristics (CMC) and Mean Average Precision (mAP) as metrics to evaluate the cross-modality retrieval performance of the proposed STHF.

B. Implementation Details

All our experiments are implemented on a single RTX8000 GPU. The proposed STHF employs the ResNet50 as the backbone network, which is initialized with pre-trained weights on ImageNet. All images of the input sequence were resized to 288×144 . Furthermore, we use random cropping, random flipping [76], and Channel-Augmented Joint Learning (CAJ) [70] to enhance the input sequence. We trained our network with an initial learning rate set to $3.5e-4$ and adopted the Adam optimizer with the cosine learning rate decay strategy [77] to adjust the learning rate dynamically. Each training is run over for 200 epochs, and 150 epochs for ablation studies. We set the sequence length to 6 and the batch size to 16 (each mini-batch includes 4 IDs and each ID includes 4 video clips of 2 modalities), respectively. For the ST-HPF, we set the f_{co}^t and f_{co}^s to 2 and 10, respectively.

TABLE III
ABLATION STUDY OF THE COMPONENTS IN STHF ON HITSZ-VCM. “B”: BASELINE. “SDC”: SHALLOW DETAIL COMPENSATION. “DSR”: DEEP SEMANTIC REFINEMENT.

Component			<i>Infrared to Visible</i>		<i>Visible to Infrared</i>	
B	SDC	DSR	R@1	mAP	R@1	mAP
✓	✗	✗	64.3	47.0	67.2	48.1
✓	✓	✗	65.1(+0.8)	49.3(+2.3)	69.3(+2.1)	50.9(+2.8)
✓	✗	✓	68.1(+3.8)	52.7(+5.7)	71.9(+4.7)	54.5(+6.4)
✓	✓	✓	70.6(+6.3)	55.2(+8.2)	72.8(+5.6)	57.4(+9.3)

TABLE IV
ABLATION STUDY ON THE MP MODULE WITH SDC AND DSR. “B”: BASELINE. “SDC”: SHALLOW DETAIL COMPENSATION. “DSR”: DEEP SEMANTIC REFINEMENT. “MP”: MODALITY PURIFY.

Component			<i>Infrared to Visible</i>		<i>Visible to Infrared</i>	
B+SDC	B+DSR	MP	R@1	mAP	R@1	mAP
✓	✗	✗	64.9	47.9	67.8	49.6
✗	✓	✗	66.6	50.9	70.3	52.8
✓	✗	✓	65.1(+0.2)	49.3(+1.4)	69.3(+1.5)	50.9(+1.3)
✗	✓	✓	68.1(+1.5)	52.7(+1.8)	71.9(+1.6)	54.5(+1.7)

C. Comparison with State-of-the-Art Methods

To verify the performance of STHF, we compared it with recent state-of-the-art methods on the HITSZ-VCM and BUPTCampus datasets, including both image-based and video-based VI-ReID approaches. Note that we did not introduce additional networks to handle the temporal information of the sequence, ensuring a fair comparison with image-based cross-modality ReID methods. Specifically, the methods we compared include: Lba [68], MPANet [23], VSD [69], CAJ [70], SEFL [71], MITML [7], IBAN [9], SADSTRM [72], SAADG [10], CST [50], AlignGAN [73], DDAG [67], AGW [1], MMN [74], DEEN [27], DART [75], MITML [7], AuxNet [8].

Evaluation on HITSZ-VCM Dataset. We first evaluated the performance of the proposed STHF on the HITSZ-VCM benchmark. The comparison results are shown in Table I, we can observe that the performance of MITML, IBAN, SADSTRM, SAADG, CST, and STHF significantly surpasses that of the VI-ReID method, demonstrating both the feasibility of the VVI-ReID task and the effectiveness of these methods. Furthermore, STHF significantly improves performance compared to IBAN and CST. Specifically, in terms of mAP, STHF achieves a 7.6% improvement in the Visible-to-Infrared setting and 7.4% in the Infrared-to-Visible setting over IBAN, while outperforming CST by 5.6% and 5.0% in the respective settings. In terms of Rank-1, STHF shows a 3.9% improvement in the Visible-to-Infrared setting and a 5.4% improvement in the Infrared-to-Visible setting over IBAN. Compared to CST, it achieves improvements of 1.0% and 0.9% in the same settings. These results further validate the effectiveness of STHF. In addition, it can be observed that STHF improves the performance by a large margin compared to IBAN, which demonstrates that our proposed sequence-level intermediate modality is more suitable for the VVI-ReID task and has been effectively exploited.

TABLE V
THE INFLUENCE OF WHICH BLOCK OF RESNET-50 TO PLUG THE SDC AND DSR MODULES. ON INFRARED TO VISIBLE RETRIEVAL.

Resnet Layer	SDC		DSR	
	R@1	mAP	R@1	mAP
Block 0	63.6	46.8	62.2	45.6
Block 1	65.8	48.9	65.5	48.7
Block 2	66.0	49.2	66.9	50.3
Block 3	65.1	49.1	68.5	52.0
Block 4	66.1	50.1	66.1	50.3

TABLE VI
ABLATION STUDY ON THE INFLUENCE OF FILTERING DIMENSION. S-HPF: SPATIAL HIGH-PASS FILTER; T-HPF: TEMPORAL HIGH-PASS FILTER; ST-HPF: SPATIAL-TEMPORAL HIGH-PASS FILTER.

Filter	<i>Infrared to Visible</i>		<i>Visible to Infrared</i>	
	R@1	mAP	R@1	mAP
S-HPF	67.5	52.1	71.9	54.7
T-HPF	67	51.9	71.3	54.6
ST-HPF	70.6	55.2	72.8	57.4

Evaluation on BUPTCampus Dataset. To verify the generalizability of the proposed STHF, we also evaluated its performance on the BUPTCampus benchmark. As shown in Table II, our method outperforms current state-of-the-art approaches with a sequence length of 6. Note that when comparing with AuxNet, for a fair comparison, we evaluate the version of AuxNet without using the re-ranking algorithm. Furthermore, compared to AuxNet, STHF significantly improves mAP and Rank-1 in the Visible-to-Infrared setting. Specifically, mAP is improved by 6.6%, while Rank-1 is increased by 8.3%, respectively. These results demonstrate the superiority of our intermediate modality compared to Fake IR [8] due to its ability to capture spatial-temporal high-frequency information.

TABLE VII

COMPARISON WITH FREQUENCY-DOMAIN METHODS. “DSR”: DEEP SEMANTIC REFINEMENT. “PHA”: PATCH WISE HIGH-FREQUENCY AUGMENTATION. “FRB”: FREQUENCY BLOCK. “ST-HPF”: SPATIAL-TEMPORAL HIGH-PASS FILTER.

Settings	Infrared to Visible		Visible to Infrared		Methods	Principle
	R@1	mAP	R@1	mAP		
Baseline (B)	64.3	47.0	67.2	48.1	-	-
B+DSR+PHA	66.6(+2.3)	50.6(+3.6)	70.3(+3.1)	52.4(+4.3)	PHA [66]	2D DHWT
B+DSR+FRB	67.2(+2.9)	51.7(+4.7)	69.9(+2.7)	53.1(+5.0)	SFMNet [58]	2D FFT
B+DSR+ST-HPF	68.1(+3.8)	52.7(+5.7)	71.9(+4.7)	54.5(+6.4)	STHF (ours)	3D FFT

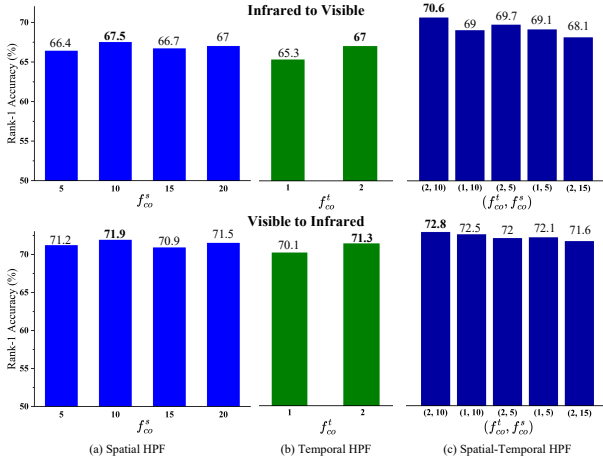


Fig. 7. Effect of cutoff frequencies f_{co}^s and f_{co}^t .

D. Ablation Study

To verify the effectiveness of each component in the proposed STHF, we conducted ablation studies on the HITSZ-VCM dataset. The experimental results are shown in Table III. The baseline method employs a two-stream ResNet50 network optimized using the identity loss (L_{ID}) and the triplet loss (L_{Tri}). STHF improves the baseline by an SDC module for detail compensation with high-frequency low-level features and a DSR module for semantic refinement with high-level features.

Effectiveness of SDC and DSR. To verify the effectiveness of SDC and DSR modules, we integrated them into the baseline sequentially. As shown in Table III, in the Infrared-to-Visible evaluation, the Rank-1 accuracy improved from 64.3% to 65.1% (+0.8%), and the mAP increased from 47.0% to 49.3% (+2.3%) due to the effect of SDC, while DSR improved the Rank-1 accuracy by 3.8% and the mAP by 5.7%, respectively. Similarly, in the Visible-to-Infrared evaluation, both SDC and DSR improved the Rank-1 accuracy and mAP to varying degrees. These results demonstrate that the SDC and DSR modules effectively enhance local details and refine global semantics. It can be observed that the DSR module contributes the most to performance improvement, which demonstrates the superiority of spatial-temporal high-frequency information in semantic information mining. Moreover, when both the SDC and DSR modules are integrated into the network, the Rank-1 accuracy increases by 6.3% and the

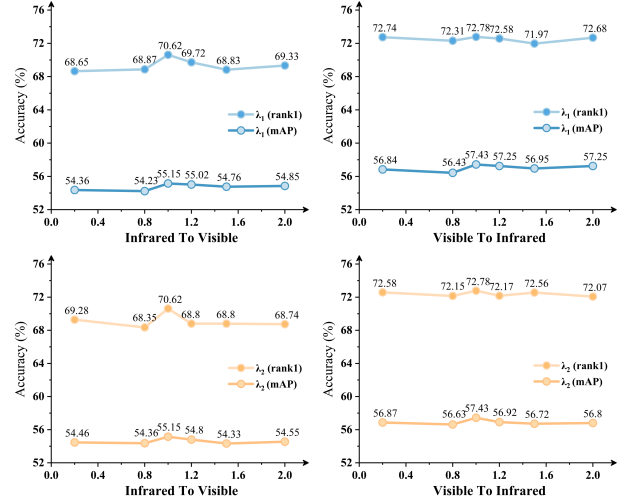


Fig. 8. Effect of the hyper-parameters λ_1 and λ_2 .

mAP improves by 8.2% in the Infrared-to-Visible evaluation, while the Rank-1 accuracy and mAP increase by 5.6% and 9.3%, respectively, in the Visible-to-Infrared evaluation. This demonstrates that the combined effect of SDC and DSR enables the model to learn comprehensive and discriminative feature representations.

Effectiveness of MP on Feature Interaction. To verify that the MP module preliminarily alleviates modality discrepancies and thereby facilitates effective feature interaction within the SDC and DSR modules, we conducted additional ablation experiments by removing MP from both components. As shown in Table IV, in the Infrared-to-Visible evaluation, introducing the MP module into the SDC leads to an improvement in Rank-1 accuracy from 64.9% to 65.1% (+0.2%) and an increase in mAP from 47.9% to 49.3% (+1.4%). Similarly, adding MP to the DSR module improves Rank-1 accuracy from 66.6% to 68.1% (+1.5%) and mAP from 50.9% to 52.7% (+1.8%). In the Visible-to-Infrared setting, the MP module similarly enhances both SDC and DSR performance in terms of Rank-1 and mAP, with improvements of 1.5%/1.3% for SDC and 1.6%/1.7% for DSR, respectively. These results confirm that eliminating style information via MP promotes more effective feature interaction and contributes to performance improvement.

Impact of Insertion Depth for SDC and DSR. To verify SDC targets low-level detail enhancement, while DSR focuses

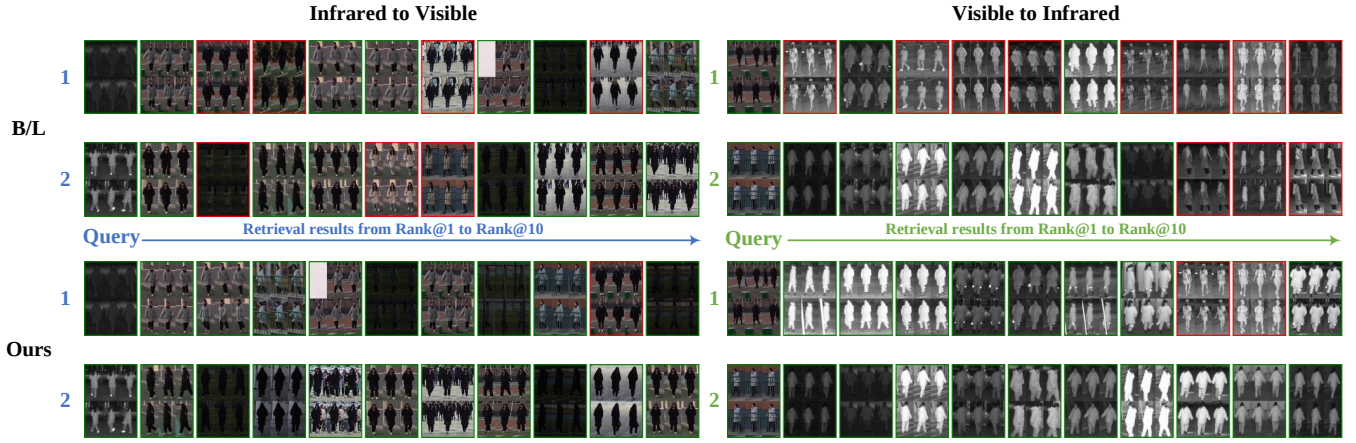


Fig. 9. Visualization of pedestrian search results, where B/L represents the baseline method.

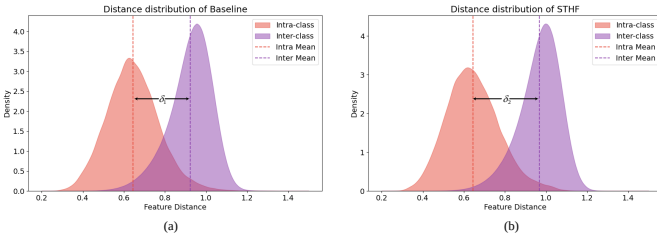


Fig. 10. Comparison of Cross-Modality Feature Distance Distributions between the baseline method and STHF.

on high-level semantic refinement, we examined the impact of integrating the SDC and DSR modules at different blocks of ResNet-50. Following common practice in representation learning [78], we consider blocks 1–2 of ResNet-50 as shallow layers that capture low-level textures and edge details, while blocks 3–4 are regarded as deep layers responsible for semantic abstraction [79]. As shown in Table V, SDC performs better when integrated into block 1 of the ResNet layers, while DSR performs better when integrated into block 3. Specifically, when applied at block 1 of the ResNet layers, in the Infrared-to-Visible evaluation, the SDC module outperforms DSR by 0.2% in Rank-1 accuracy and 0.3% in mAP. In contrast, at block 3 of the ResNet layers, the DSR module surpasses SDC by 3.4% in Rank-1 accuracy and 2.9% in mAP. These results demonstrate that low-level features contain detailed information suitable for information compensation, while high-level features extract semantic category information suitable for information guidance, aligning with our analysis in the previous sections.

Influence of Filtering Strategy. To verify the effectiveness of spatial-temporal high-pass filtering in the VVI-ReID task, we conducted ablation experiments by constructing intermediate modalities using different filtering strategies. The cutoff frequencies were set to $f_{co}^s = 10$ for spatial filtering and $f_{co}^t = 2$ for temporal filtering. As shown in Table VI, in the Infrared-to-Visible setting, ST-HPF outperforms S-HPF and T-HPF by 3.1% and 3.6% in Rank-1 accuracy, and by 3.1% and 3.3% in mAP, respectively. A similar trend is observed in the Visible-

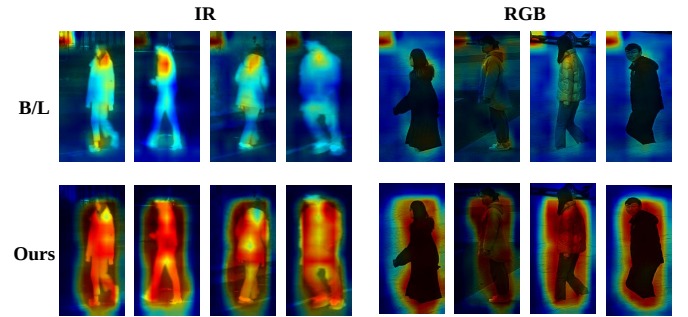


Fig. 11. Comparison of CAM between the baseline method and STHF.

to-Infrared setting, where ST-HPF achieves improvements of 0.9% and 1.5% in Rank-1, and 2.7% and 2.8% in mAP over S-HPF and T-HPF, respectively. These results confirm that spatial-temporal filtering captures complementary frequency cues, enabling more discriminative intermediate representations than single-dimensional approaches.

Comparison of different frequency-domain methods. To evaluate the impact of different frequency learning strategies, we replaced ST-HPF with Patch wise High-frequency Augmentation (PHA) [66] and Frequency Block (FRB) [58] while leveraging the DSR module to learn frequency-domain information. Specifically, we directly replace the original ST-HPF with PHA, using the HFC-enhanced sequence as the intermediate modality. As for FRB, we utilize global frequency-domain features to replace the intermediate modality representations, enabling spatial-frequency mutual learning. As shown in Table VII, FRB achieves better average performance than PHA, indicating that FFT-based frequency modeling provides a more complete and expressive frequency representation than DHWT. Furthermore, ST-HPF integrates 3D FFT to jointly model HFC in both spatial and temporal dimensions, achieving the best results across all metrics. These results demonstrate the effectiveness of spatial-temporal high-frequency modeling in enhancing feature representation and alleviating the modality discrepancy in the VVI-ReID task.

E. Parameter Analysis

The proposed STHF involves four key hyperparameters: f_{co}^t , f_{co}^s , λ_1 , and λ_2 . Specifically, f_{co}^t and f_{co}^s control the filtering intensity of the spatial-temporal high-pass filter (ST-HPF), while λ_1 and λ_2 balance the loss contribution of the intermediate modality. To investigate their impact on the overall performance of the framework, we conduct corresponding experiments and hyperparameter analysis.

The cutoff frequencies f_{co}^t and f_{co}^s determine the HFC retained in the spatial-temporal spectrum, directly influencing the preservation of spatial and temporal details in intermediate modalities. According to Section III-C, the valid ranges of these frequencies are defined as follows: the temporal cutoff frequency f_{co}^t varies within $[1, T/2)$, and the spatial cutoff frequency f_{co}^s lies in the range $[1, \min(H/2, W/2)]$. To analyze their impact, we evaluate the performance of STHF under three different configurations, as illustrated in Fig. 7.

Effect of f_{co}^s in S-HPF. We first investigate how different spatial cutoff frequencies affect performance by varying $f_{co}^s \in \{5, 10, 15, 20\}$. As shown in Fig. 7(a), the best result is obtained when $f_{co}^s = 10$, indicating that moderate spatial filtering effectively preserves important details while reducing noise.

Effect of f_{co}^t in T-HPF. We then examine the effect of temporal cutoff frequencies, with $f_{co}^t \in \{1, 2\}$. As shown in Fig. 7(b), the model achieves peak performance when $f_{co}^t = 2$, suggesting that slightly stronger temporal filtering improves dynamic feature modeling.

Joint effect of f_{co}^s and f_{co}^t in ST-HPF. Finally, we explore joint spatial-temporal configurations, starting from the best-performing combination ($f_{co}^s = 10$, $f_{co}^t = 2$), and testing variants with stronger or weaker filtering. As shown in Fig. 7(c), the strongest filtering setting ($f_{co}^s = 15$, $f_{co}^t = 2$) results in significant performance degradation, while the original setting maintains the highest accuracy. These results demonstrate that weak filtering may provide limited discriminative cues, while excessive filtering could suppress essential structural semantics. Based on these observations, we empirically set $f_{co}^s = 10$ and $f_{co}^t = 2$ for ST-HPF to ensure balanced spatial-temporal HFC preservation.

The hyperparameter λ_1 controls the contribution of the cross-entropy loss from the intermediate modality, which promotes identity classification by supervising the predicted probability distributions. As shown in Fig. 8, we evaluate the performance of STHF across different values of λ_1 within the range $[0.2, 2]$. According to the results, we empirically set $\lambda_1 = 1$ for optimal performance.

The hyperparameter λ_2 balances the influence of the triplet loss from the intermediate modality. This loss optimizes the feature space through metric learning to enhance feature discriminability. As shown in Fig. 8, we report the performance of STHF under different parameter configurations of λ_2 , within the same range as λ_1 . Based on the experiment results, we fix $\lambda_2 = 1$.

F. Visualization

To demonstrate the effectiveness of the proposed STHF, we visualized and compared the retrieval results, cross-modality

feature distance distributions, and feature focus region between the baseline method and STHF.

Retrieval Results. We visualized the ReID results of the baseline method and STHF on the Infrared-to-Visible and Visible-to-Infrared pedestrian retrieval tasks using the HITSZ-VCM dataset. As shown in Fig. 9, given a query pedestrian sequence, we retrieved the top 10 most similar pedestrian sequences, where the green boxes indicate the correct retrieved results and the red boxes denote incorrect ones. The visualization results show that the proposed STHF outperforms the baseline method in complex scenarios. For example, in the Infrared-to-Visible retrieval, Query 1 has low visibility, and Query 2 is affected by occlusion in the Visible-to-Infrared retrieval, the performance of the baseline method is impacted by these factors, whereas STHF exhibits stronger robustness.

Feature Distance Distributions. We visualized the feature distance distributions of the intra-class and inter-class samples between the baseline method and STHF as shown in Fig. 10(a) and Fig. 10(b). It can be distinctly observed that the mean distance between the intra-class and inter-class of STHF is larger than that of the baseline method (i.e., $\delta_2 > \delta_1$), indicating that spatial-temporal high-frequency information effectively improves the discriminability between different classes.

Feature Focus Region. We compared class activation maps (CAMs) [80] between the baseline method and STHF. As shown in Fig. 11, the proposed STHF effectively mitigates the influence of static backgrounds and objects. Specifically, STHF captures features that focus on both the overall structure and fine details of pedestrians, whereas the baseline method tends to focus only on a specific part of the pedestrian and is more susceptible to background interference.

V. CONCLUSION

In this paper, we propose a novel framework named STHF, which extracts discriminative spatial-temporal cues while mitigating modality differences from a frequency domain perspective. Specifically, we propose a novel sequence-level intermediate modality based on 3D FFT: Spatial-Temporal High-Frequency Information, which enhances structural and motion-based identity cues. Furthermore, we design two feature interaction modules: a Shallow Detail Compensation (SDC) module in shallow layers to enhance local details, and a Deep Semantic Refinement (DSR) module in deep layers to refine global representation. Experimental results on the public HITSZ-VCM and BUPTCampus datasets demonstrate the effectiveness and superiority of the proposed STHF.

Moreover, although Spatial-Temporal High-Frequency Information mitigates modality discrepancy, a fixed cutoff frequency introduces redundancy for some identities and over-filtering for others, reducing discriminability. Besides, STHF incurs high training costs due to an additional network to learn high-frequency information. In future work, we aim to develop a method that can adaptively select the appropriate cutoff frequency for each identity to achieve better filtering performance.

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